



Outdoor Field Testing

Range Test Report

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Preface

This document presents the findings and analysis of the outdoor field-testing range conducted to evaluate the signal range and performance of the tested system in an outdoor environment.

Privacy Information

This document may contain information of a sensitive nature. This information should not be given to any persons outside of the Avartanam test labs LLP. without prior consent.

Report Prepared by

Name	Position	Signature	Date
Bhargen Shah	RF Engineer		09/08/2023

Approval

Name	Position	Signature	Date
Himanshu Ankleshawaria	Senior Manager	Email Approval	10/8/2023

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1.Introduction

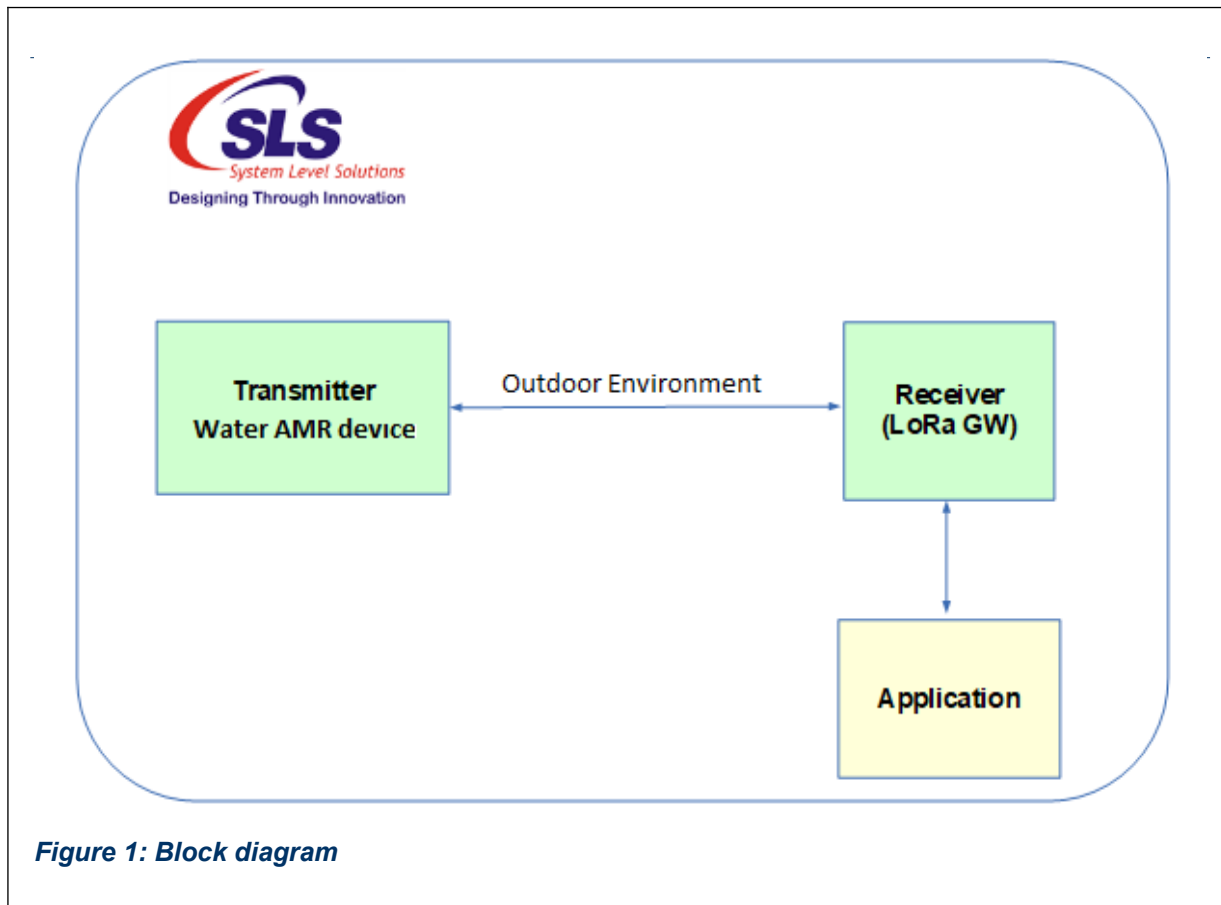
- This document describes the information about Out-Door RF Range Testing of Water AMR devices at installation location Jaipur.
- When selecting right Radio Technology, the achievable communication range is an important factor.

1.1 Why Range Test Performed?

- Range tests are performed to determine the maximum distance or extent over which a particular device, system, or object can operate effectively. They help evaluate the performance, reliability, and limitations of the subject under consideration within a specific range of operation.

2. Block diagram

- The block diagram represents a visual depiction of the setup for transmitting and receiving signals in an urban environment. It provides a graphical representation of the components and connections involved in the transmission and reception process, allowing for a better understanding of the system's configuration and operation.



- As above block diagram shows graphical representation of Transmitter and Receiver setup in Customer site location.

3. Range Testing with Standard Proven Handy Tool

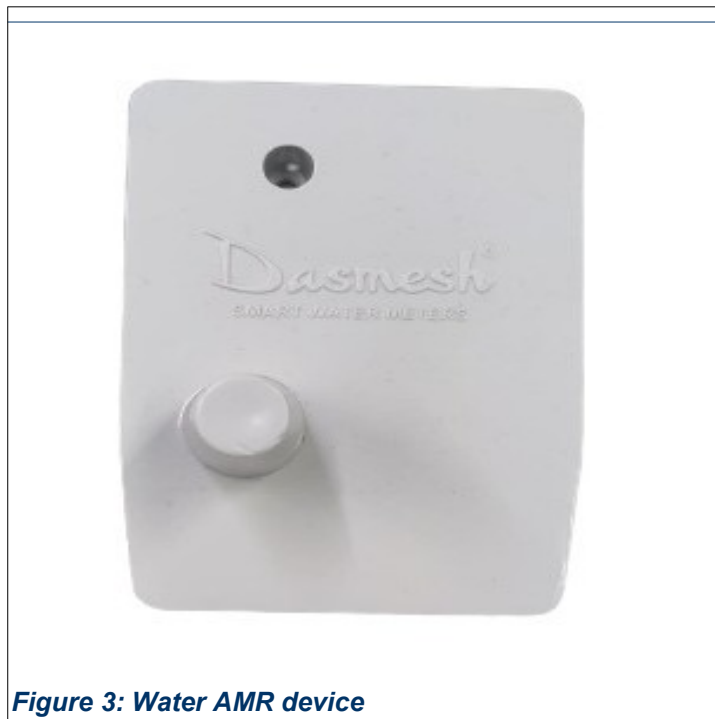
- **RF Power Meters:** These tools measure the power level of RF signals, allowing you to assess signal strength and determine the range of a transmitter.
- **Spectrum Analyzers:** Spectrum analyzers provide a visual representation of the frequency spectrum, enabling you to identify and analyze RF signals present in the environment. They help in detecting interference and optimizing frequency usage.
 - **Noise Level measurement:** The noise level in an RF survey refers to the amount of unwanted electromagnetic signals or interference present in the radio frequency environment. It is an important factor to consider when evaluating the quality and reliability of wireless communication systems, as high levels of noise can degrade signal clarity and reduce the overall performance of RF devices.
- **Field Strength Meters:** These devices measure the strength of RF signals at specific locations, allowing you to map out signal coverage and identify areas with weak or strong signal reception.
- **Signal Generators:** Signal generators produce RF signals of known frequency and power, which can be used to test the sensitivity and range of receivers.

4. Site location

- **Site Location:** Jaipur
- **Lat/Long data:** 26.918548,75.823238



5. Water AMR device



LoRa Module Device Configuration:

- Module: NLN500A
- Output Power: +21 dBm
- Operating Frequency: 865MHz to 868MHz
- Spreading Factor: SF9
- Coding Rate: 4/8
- Transmit Time: 1 minute

* Range test is performed with power level of +21 dBm.

6.NLG7644 (LoRa Gateway)

- Gateway location optimization refers to the process of strategically determining the optimal placement of gateways in a network infrastructure. This involves selecting suitable locations that maximize network coverage, minimize signal interference, and ensure efficient communication between devices and the gateway. By optimizing gateway locations, network performance can be enhanced, ensuring reliable connectivity and effective data transmission throughout the network.



Figure 4: NLG7644 LoRa Gateway location 1 at customer site

7.Site analysis

- During the Jaipur RF site survey, customer had installed around 210 water AMR meters with one NLG7644 (LoRa gateway).
- Location of installation the NLG7644 (LoRa gateway) on the corner side (figure 5) of site without maintaining the antenna height from the ground.
- Each house having a separate water supply pipeline with one smart AMR meter was installed inside or outside the house area.
- In some house water meter was open towards sky which gave the good connectivity, but other houses water meters were surrounded or inside concrete cabin and closed from top by concrete cabin.
- Also, some water meters were placed on ground and not maintained 3-feet height above the ground that may also degrade the RF performance (figure 6).
- There are more than one GSM towers near the buildings which continuously producing the out-band noise.



Figure 5: Customer gateway location



Figure 6: Improper placement of Water AMR devices (not maintained the height from the ground)

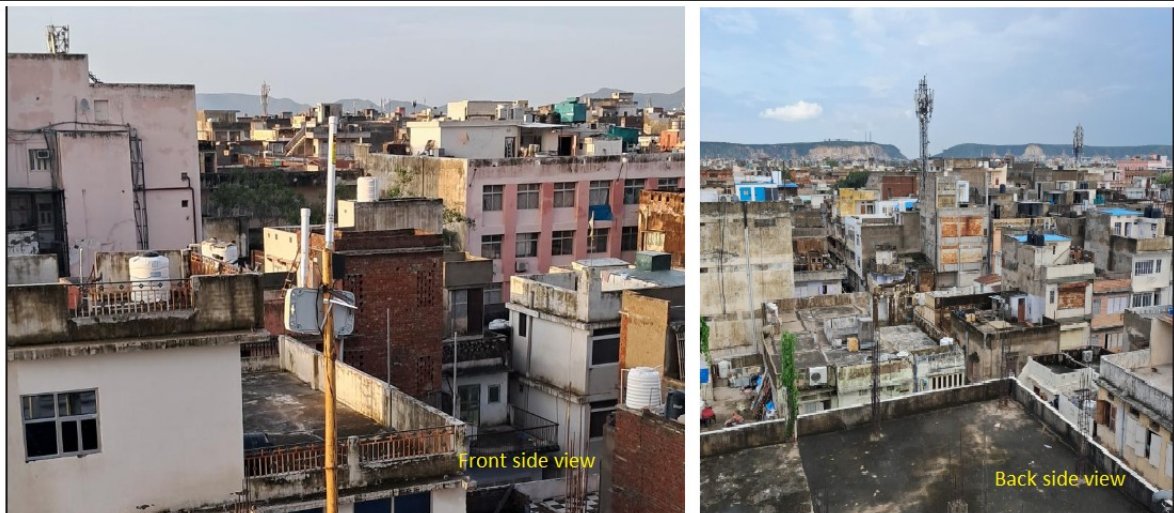
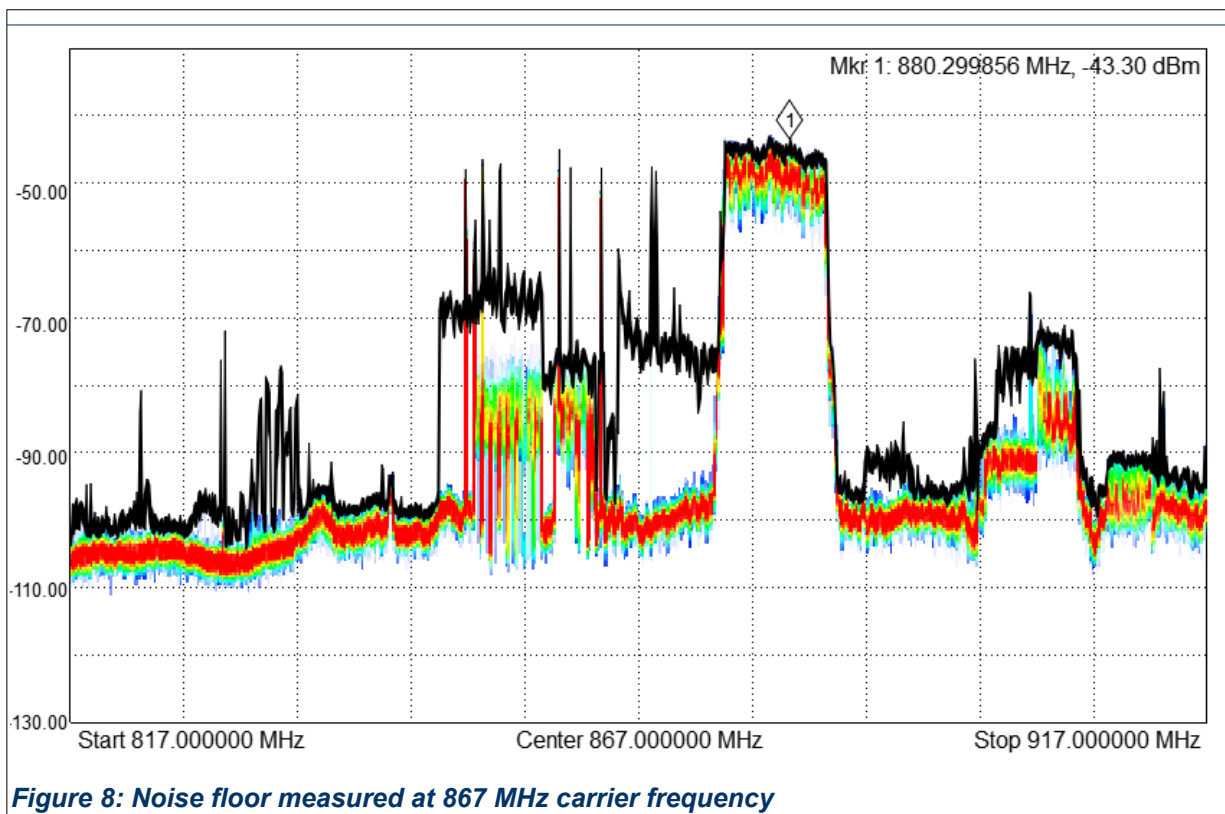


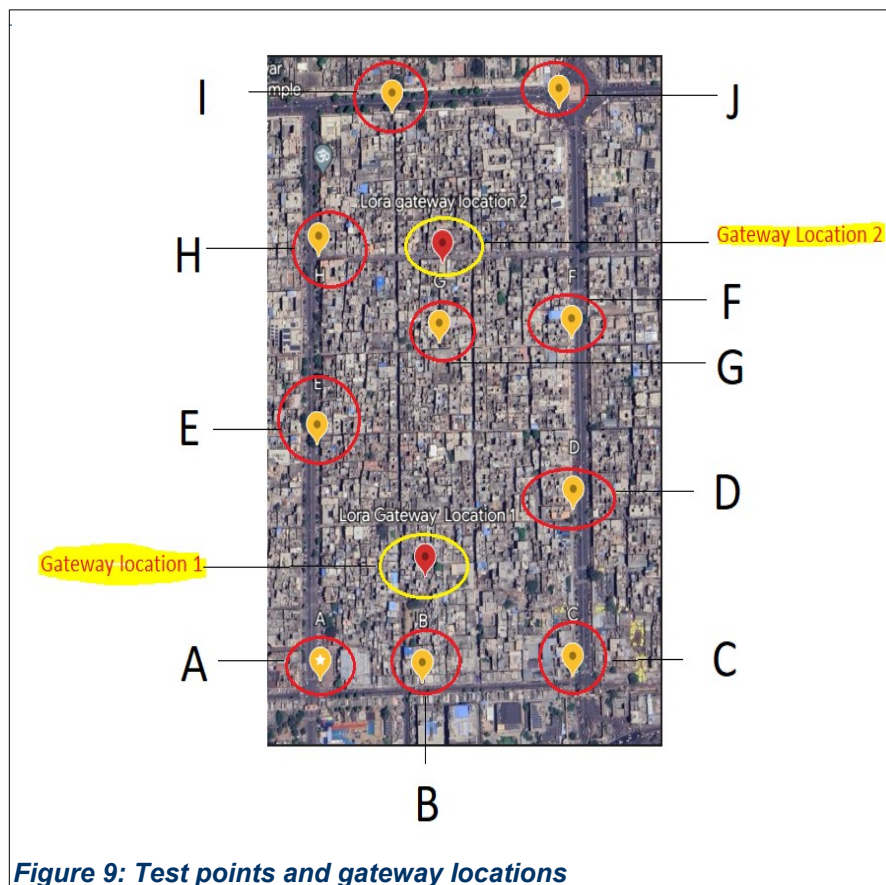
Figure 7: Final location of LoRa gateway 1 at Jaipur

8. Test Procedure

- To investigate the presence of interference in the LoRaWAN networks, we conducted a comprehensive interference-hunting campaign using specialized equipment such as spectrum analyzer, Avartanam filter & signal hound.
- We collected empirical data by taking measurements at various locations and frequencies and analyzed the data to identify the sources of interference.
- We also identified such a place for LoRa gateway on the Jaipur customer site which is located on center has been selected after by observing the noise and installation parameters.



- During the RF site survey, we placed the only one LoRa gateway to cover few test points (A to F) water AMR devices locations and for second LoRa gateway customer faced the permission related issue during the testing time So we haven't carried out remaining test points (G to J).
- Afterward our team placed the Water AMR device in randomly on the Different location on customer site some was tested on the open-air environments, and some devices tested with already installed water AMR devices.
- The AMR device is running on the battery supply, on every 1 min device sent the uplink message and checked the RSSI value on the AMR debug portal/website for each location.
- Our team carried out Range test at different location at Jaipur site inside and outside the residence house. Majority tests cases was performed inside the house where the actual meters were installed.



9.Measurement Results

Gateway Location: NLG7644 LoRa Gateway 1 (26.918548,75.823238)		
Node Location: Test Point A		
Data Rate	RSSI	Packet success ratio (%)
SF9	-109	100
	-112	
	-112	
Node Location: Test Point B		
Data Rate		Packet success ratio (%)
SF9	-106	100
	-103	
	-106	
Node Location: Test Point C		
Data Rate		Packet success ratio (%)
SF9	-108.2	100
	-110	
	-105.2	
Node Location: Test Point D		
Data Rate		Packet success ratio (%)
SF9	-109.5	100
	-110	
	-108	
Node Location: Test Point E		
Data Rate		Packet success ratio (%)
SF9	-90.5	100
	-103	
	-94	
Node Location: Test Point F		
Data Rate		Packet success ratio (%)
SF9	-113	100
	-111	
	-109	

Table 1: Measurement Test results

- There will be variation in the RSSI value due to the bad weather condition or any object passing between the AMR & Gateway path.
- For the inside house or on the ground water AMR devices (figure 6) the packet success ratio was only 50%.

10. Suggestions

Our team has thoroughly studied a water meter location at Jaipur customer site and carefully understood the requirement. The LoRa Gateway required sizeable space that would be safe and secure.

- Increase height of the Water AMR device at least 3 feet above from the ground level for better RF Range performance.
- Both the gateways must require to be install with Avartanam filter on mentioned location in Figure 9.
- During the testing remaining test point from G to J AMR devices was not connected with the Gateway number 1. So, for that Customer must have to install 2nd gateway on the suggested locations to connect the test point G to J water AMR devices.
- Water AMR devices that already installed in concrete cabin, inside metal housing, underground require to be configured on SF12 to increase the performance or need to change the height of that AMR meters for the long-range communications.
- Alone Battery is not delivering enough Power (+21dBm) so Battery with Super-Cap is required. Avartanam LAB Engineers have cross verified performance of both cases with and without super-Cap. With super-cap AMR devices has started work and connect with the LoRa gateway.
- Battery with super-cap will also improve the RF range performance.
- Put Lighting Arrester with the LoRa gateway for Preventing Product Damage against over-voltage and surge current induced by bolts of lightning.